

Unit Functional Design Specification – Enzymatic Interesterification

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5.8	Named Set	Error! Bookmark not defined.

The following table describes the EM classes that shall be used along with their instance configuration.

EM Class	Sequence Name
EM_EIE_STARTUP	STARTUP E2600
EM_EIE_FEED	E2682A
EM_EIE_REACTOR	E2603A
EM_EIE_REACTOR	E2603B
EM_EIE_REACTOR	E2603C
EM_EIE_REACTOR	E2603D
EM_EIE_PRODUCT1	E2635E
EM_EIE_PRODUCT2	E2682B

1 EQUIPMENT MODULE CLASS DESIGN – EM_EIE_STARTUP (STARTUP E2600)

1.1 Functional Description

EM_EIE_STARTUP is a class based equipment module which starts the Auxiliary Circuits and initiates production process. EM_EIE_STARTUP is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Aux On Command and should be released in Aux Off Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

1.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_EIE_STARTUP	EI1EQ_2600E
DEV1	_V12_IP_130_SE	EI1_V1150FW
DEV2	_MRSS11_130_SE	EI1_PU115003
DEV3	_CSD_130_SE	EI1_TT1121W1
DEV4	EM_EIE_FEED	EI1EQ_2682A
DEV5	EM_EIE_PRODUCT1	EI1EQ_2635E
DEV6	EM_EIE_REACTOR	EI1EQ_2603A
DEV7	EM_EIE_REACTOR	EI1EQ_2603B
DEV8	EM_EIE_REACTOR	EI1EQ_2603C
DEV9	EM_EIE_REACTOR	EI1EQ_2603D
DEV10	EM_EIE_PRODUCT2	EI1EQ_2682B1
DEV11	EM_EIE_PRODUCT2	EI1EQ_2682B2

Table 1: Instances

1.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

1.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Set DEV3 Mode to Remote Output
		A0040	Set DEV3 output to Closed and Disable Conditional Logic
T9900A	Actions Complete or Time Out & HOLD=TRUE		Go to step H9900A
T9900B	Actions Complete or Time Out & HOLD=FALSE		Go to step H9900B
H9900A	Hold Complete	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
END1	Command Complete		True

Table 2: Hold Sequence

1.3.2 Command 01 – Aux On

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux On Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initiliazations	A0010	First Pass Logic
		A0020	First Pass = True & Set Task Pointer
		A0030	Acquire EM
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Device	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Reset HOLD_REQ
T0100	Standby		Go to step S0100
S0100	Standby	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Set DEV3 Mode to Remote Output
		A0040	Set DEV3 output to Closed and Disable Conditional Logic
T0200	Start Aux		If <AUX> (OP004) latch is ON then go to step S0200
S0200	Start Aux	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Setting DEV3 setpoint to (TT1121W1 SP) and Enabling Conditional Logic
T9900	Complete		If Auxiliary circuits running and <HOLD> (OP003) latch OFF and <RUN> (OP001) latch ON then go to step S9900



Step	Description	Action	Action Description
S9900	Complete	A0010	Set Message: Aux On Command Complete

Table 3: Aux On Sequence

1.3.3 Command 02 – Aux Off

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux Off Command
T0100	Aux Off		Go to step S0100
S0100	Aux Off	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Stopping and Disabling Conditional Logic for DEV2
		A0030	Set DEV3 Mode to Remote Out
		A0040	Set DEV3 output to Closed and Disable Conditional Logic
		A0050	Reset OP parameters
T0200	Release Devices		If <AUX> (OP004) latch is OFF then go to step S0200
S0200	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
T9900	Complete		If no pending actions in S0200 then go to step S9900
S9900	Complete	A0010	Set Message: Aux Off Complete
END	Command Complete		True

Table 4: Aux Off Sequence

1.3.4 Command 03 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0100	Production		If Task Pointer = 1 then go to step S0100
T0200A	Drain		If Task Pointer = 2 then go to step S0200
S0100	Production	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Setting DEV3 setpoint to (TT1121W1 SP) and Enabling Conditional Logic
		A0040	Acquiring DEV4
		A0050	Acquiring DEV5
		A0060	Acquiring DEV6
		A0070	Acquiring DEV7
		A0080	Acquiring DEV8
		A0090	Acquiring DEV9
		A0100	Acquiring DEV10
		A0110	Set Fill Command for DEV10
		A0120	Acquiring DEV11
		A0130	Set Fill Command for DEV11
T0200B	Drain		If <HOLD> (OP003) latch is OFF and <DRAIN> (OP002) latch is ON then go to step S0200
S0200	Drain	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Setting DEV3 setpoint to (TT1121W1 SP) and Enabling Conditional Logic
		A0040	Set Message: Executing Drain
		A0050	Set Drain Command for DEV4
		A0060	Set Command DEV5
		A0070	Set Command DEV6
		A0080	Set Command DEV7
		A0090	Set Command DEV8
		A0100	Set Command DEV9
		A0110	Acquiring DEV10
		A0120	Set Fill Command for DEV10



Step	Description	Action	Action Description
		A0130	Acquiring DEV11
		A0140	Set Fill Command for DEV11
		A0150	Reset Production command
T9900	Complete		If E2682A is in step <Standby> and all E2603x are in step <Clean> or <Standby> and all E2682Bx are in step <Standby> then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
END	Command Complete		True

Table 5: Production Sequence

1.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	RUN	Named Set	NA	OFF
OP002	DRAIN	Named Set	NA	OFF
OP003	HOLD	Named Set	NA	OFF
OP004	AUX	Named Set	NA	OFF
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	TT1121W1 SP	Floating Point	NA	20

Table 6: Operating and Tuning Parameters

1.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		E11EQ_2600E
OPR_CMD_MASK		15
LSL_150	External Reference	E11_LSL1150/DI1/DISC_ACT

Table 7: Configurable Parameters



1.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_EIE_STARTUP	Enzymatic Interesterification Area: Startup EM

Table 8: Named Set Details

Details of states for NS_EIE_STARTUP named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Aux On
02	Aux Off
03	Production

Table 9: Name Set State Details

2 EQUIPMENT MODULE CLASS DESIGN – EM_EIE_FEED (E2682A)

2.1 Functional Description

EM_EIE_FEED is a class based equipment module that performs filling operation followed by Draining and Blowing in order to complete Enzymatic Interesterification. EM_EIE_FEED is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Production Command and should be released on completion of Drain Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

2.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_EIE_FEED	EI1EQ_2682A
DEV1	_V12_IP_130_SE	EI1_V1082A
DEV2	_V12_IP_130_SE	EI1_V1082AD
DEV3	_V12_IP_130_SE	EI1_V1082AF
DEV4	_V12_IP_130_SE	EI1_V1082AV
DEV5	_V12_IP_130_SE	EI1_V1082AR
DEV6	_MRSS11_130_SE	EI1_PU1034NA
DEV7	_MRSS11_130_SE	EI1_PU105081A
DEV8	_MVS11_IL_130_SO	EI1_PU1082A
DEV9	_CSD_130_SE	EI1_TT10211
DEV10	_CMT_130_SE	EI1_TT1081A1
DEV11	_CMT_130_SE	EI1_FT1034NA
DEV12	_CMT_130_SE	EI1_FT1082A1

Table 10: Instances

2.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

2.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A0010	Store Fail Message
		A0020	Reset HOLD_REQ
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
		A0090	Set DEV9 output to Closed and Disable Conditional Logic
		A0100	Set DEV10 output to Closed and Disable Conditional Logic
		A0110	Set DEV11 output to Closed and Disable Conditional Logic
		A0120	Set DEV112 output to Closed and Disable Conditional Logic
T9900A	Actions Complete or Time Out & HOLD=TRUE		Go to step H9900A

Step	Description	Action	Action Description
T9900B	Actions Complete or Time Out & HOLD=FALSE		Go to step H9900B
H9900A	Hold Complete	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
END1	Command Complete		True

Table 11: Hold Sequence

2.3.2 Command 01 – Production

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Production Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True

Step	Description	Action	Action Description
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Acquire Private Devices
		A0120	Acquire Private Devices
		A0130	Acquire Private Devices
		A0140	Set Hold Request
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Production		If Task Pointer = 3 then go to step S0300
S0100	Standby	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Closed and Disable Conditional Logic



Step	Description	Action	Action Description
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
		A0150	Set DEV12 Mode to Remote Out
		A0160	Set DEV12 output to Closed and Disable Conditional Logic
T0200A	Fill		If no pending actions in S0100 then go to step S0200
S0200	Fill	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Closed and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Cascade
		A0140	Setting DEV11 setpoint to (FT1034NA RSP) and Enabling Conditional Logic
		A0150	Set DEV12 Mode to Remote Output
		A0160	Set DEV12 output to Closed and Disable Conditional Logic
		A0170	Waiting for LT1082A1 > TP003
		A0180	Set Task Pointer

Step	Description	Action	Action Description
T0300A	Production		If LT1082A1 >= Production Level SP (TP003) then go to step S0300
S0300	Production	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Cascade
		A0100	Setting DEV9 setpoint to (TT10211 RSP) and Enabling Conditional Logic
		A0110	Set DEV10 Mode to Remote Cascade
		A0120	Setting DEV1 setpoint to (TT1081A RSP) and Enabling Conditional Logic
		A0130	Set DEV11 Mode to Remote Cascade
		A0140	Setting DEV11 setpoint to (FT1034NA RSP) and Enabling Conditional Logic
		A0150	Set DEV12 Mode to Remote Cascade
		A0160	Setting DEV1 setpoint to (FT1082A1 RSP) and Enabling Conditional Logic
		A0170	Set Task Pointer
T9900	Complete		If no pending actions in S0300 then go to step S9900
S9900	Complete	A0010	Set Message: Production Command Complete
END	Command Complete		True

Table 12: Production Sequence

2.3.3

Command 02 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
T0100	Drain		If Task Pointer = 3 then go to step S0100

Step	Description	Action	Action Description
T0200B	Blow		If Task Pointer = 4 then go to step S0200
T0300B	Standby		If Task Pointer = 5 then go to step S0300
S0100	Drain	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Cascade
		A0100	Setting DEV9 setpoint to (TP007_DESC) and Enabling Conditional Logic
		A0110	Set DEV10 Mode to Remote Cascade
		A0120	Setting DEV10 setpoint to (TP004_DESC) and Enabling Conditional Logic
		A0130	Set DEV11 Mode to Remote Output
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
		A0150	Set DEV12 Mode to Remote Cascade
		A0160	Setting DEV12 setpoint to (TP006_DESC) and Enabling Conditional Logic
T0200A	Blow		If LT1082A1 in VLL alarm and LSL_082A1 in alarm then go to step S0200
S0200	Blow	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6

Step	Description	Action	Action Description
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Closed and Disable Conditional Logic
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
		A0150	Set DEV12 Mode to Remote Out
		A0160	Set DEV12 output to Closed and Disable Conditional Logic
		A0170	Set Blower Time
		A0180	Start Blower Time
		A0190	Blower Time Message
		A0200	Set Task Pointer
T0300A	Standby		If Blow Timer (TP002) has elapsed then go to step S0300
S0300	Standby	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Stopping and Disabling Conditional Logic for DEV6
		A0070	Stopping and Disabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
		A0090	Set DEV9 Mode to Remote Out
		A0100	Set DEV9 output to Closed and Disable Conditional Logic



Step	Description	Action	Action Description
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
		A0150	Set DEV12 Mode to Remote Out
		A0160	Set DEV12 output to Closed and Disable Conditional Logic
		A0170	Set Task Pointer
T0400	Disable HM and SM		If no pending actions in S0300 then go to step S0400
S0400	Disable HM and SM	A0010	Disable HM 01 (Module Ownership Lost)
T0500	Release Devices		If no pending action in S0400 then go to step S0500
S0500	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
T9900	Complete		If no pending actions in S0500 then go to step S9900
S9900	Complete	A0010	Set Message: Drain Command Complete
END	Command Complete		True

Table 13: Drain Sequence



2.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Caustic Dosing	Named Set	N/A	OFF
OP002	Recirculation	Named Set	N/A	OFF
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Blow Timer	Floating Point	sec	500
TP003	Production Level SP	Floating Point	%	0

Table 14: Operating and Tuning Parameters

2.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG	String	EI1EQ_2682A
OPR_CMD_MASK	32 bit unsigned integer	7
DY_REF_1	Dynamic Reference	
LT1082A1	String	EI1_LT1082A1
LSL_082A1	External Reference	EI1_LSL1082A1/DI1/DISC_ACT
HOLD	External Reference	EI1EQ_2600E/OP003_VALUE
STEP_TIME	Floating Point	
COMPLETED_TIME	Floating Point	
RESET_COMP_TM	Boolean	

Table 15: Configurable Parameters

2.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_EIE_FEED	Enzymatic Interesterification Area: Feed

Table 16: Named Set Details

Details of states for NS_EIE_FEED named set are as follows.

Named Set Value	Named Set Name
00	Hold
01	Production
02	Drain

Table 17: Name Set State Details

3 EQUIPMENT MODULE CLASS DESIGN – EM_EIE_REACTOR (E2603X)

3.1 Functional Description

EM_EIE_REACTOR is a class based equipment module that checks the availability of enzyme and accordingly executes EIE process for oil and enzyme. EM_EIE_REACTOR is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Process Fluid Command and should be released on completion of Blow Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

3.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name			
EM	EM_EIE_REACTOR	EI1EQ_2603A	EI1EQ_2603B	EI1EQ_2603C	EI1EQ_2603D
DEV1	_V12_IP_130_SE	EI1_V1103ABN	EI1_V1103BBN	EI1_V1103CBN	EI1_V1103DBN
DEV2	_V12_IP_130_SE	EI1_V1103AD1	EI1_V1103BD1	EI1_V1103CD1	EI1_V1103DD1
DEV3	_V12_IP_130_SE	EI1_V1103AD2	EI1_V1103BD2	EI1_V1103CD2	EI1_V1103DD2
DEV4	_V12_IP_130_SE	EI1_V1103AF1	EI1_V1103BF1	EI1_V1103CF1	EI1_V1103DF1
DEV5	_V12_IP_130_SE	EI1_V1103AF2	EI1_V1103BF2	EI1_V1103CF2	EI1_V1103DF2
DEV6	_V12_IP_130_SE		EI1_V1193BFY	EI1_V1103CFY	
DEV7	_V12_IP_130_SE	EI1_V1103AQE	EI1_V1103BQE	EI1_V1103CQE	EI1_V1103DQE
DEV8	_V12_IP_130_SE	EI1_V1103AR	EI1_V1103BR	EI1_V1103CR	EI1_V1103DR
DEV9	_V12_IP_130_SE	EI1_V1103AV	EI1_V1103BV	EI1_V1103CV	EI1_V1103DV

Table 18: Instances

3.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

3.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
T9900A	Actions Complete or Time Out & HOLD=TRUE		Go to step H9900A
T9900B	Actions Complete or Time Out & HOLD=FALSE		Go to step H9900B
H9900A	Common Hold End	A0010	Completed

Step	Description	Action	Action Description
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
END1	Command Complete		True

Table 19: Hold Sequence

3.3.2 Command 01 – Process Fluid

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Process Fluid Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices

Step	Description	Action	Action Description
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Acquire Private Devices
		A0120	Reset HOLD_REQ
T0100	Standby		Go to step S0100
T0200B	Fill Enzyme		If Task Pointer = 2 then go to step S0200
T0300B	Conditioning		If Task Pointer = 3 then go to step S0300
T0400B	Drain E2635E		If Task Pointer = 4 then go to step S0400
T0500B	Blow PE2635E		If Task Pointer = 5 then go to step S0500
T0600D	Ready		If Task Pointer = 6 then go to step S0600
T0700B	Fill Oil		If Task Pointer = 7 then go to step S0700
T0800B	EIE		If Task Pointer = 8 then go to step S0800
T0900B	Blow		If Task Pointer = 9 then go to step S0900
T1000B	Clean		If Task Pointer = 10 then go to step S1000
T1100C	Close valves		If Task Pointer = 11 then go to step S1100
S0100	Standby	A0020	Set DEV1 Mode to Cascade
		A0030	Set DEV2 Mode to Cascade
		A0040	Set DEV3 Mode to Cascade
		A0050	Set DEV4 Mode to Cascade
		A0060	Set DEV5 Mode to Cascade
		A0070	Set DEV6 Mode to Cascade
		A0080	Set DEV7 Mode to Cascade
		A0090	Set DEV8 Mode to Cascade
		A0100	Set DEV9 Mode to Cascade
		A0110	Closing and Disabling Conditional Logic for DEV1
		A0120	Closing DEV2
		A0130	Closing and Disabling Conditional Logic for DEV3
		A0140	Closing and Disabling Conditional Logic for DEV4
		A0150	Closing and Disabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0160	Closing and Disabling Conditional Logic for DEV6
		A0170	Closing and Disabling Conditional Logic for DEV7
		A0180	Closing DEV8
		A0190	Closing and Disabling Conditional Logic for DEV9
		A0200	Set Message: Waiting for EM_EIE_PRODUCT1 not to be in step Blow
T0200A	Fill Enzyme		If (STARTUP is in step <Production> or <Drain>) and E2603x <Enzyme Empty> ON and PRODUCT1 is in step <Ready> and push button <Start> (OP002) then go to step S0200
T0600B	Ready		If (STARTUP is in step <Production> or <Drain>) and E2603x <Enzyme Present> ON and push button <Start> (OP002) then go to step S0600
S0200	Fill Enzyme	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Waiting for LALL1103X not to be in alarm
		A0110	Set Delay Time
		A0120	Start Delay Time
		A0130	Delay Time Message
		A0140	Set Task Pointer and Enzymes
T0300A	Conditioning		If LSL1103X is not in alarm then go to step S0300 after 120 sec delay

Step	Description	Action	Action Description
S0300	Conditioning	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Set ConditioningTime
		A0110	Start Conditioning Time
		A0120	Conditioning Time Message
		A0130	Set Task Pointer
T0400A	Drain E2635E		If MPE2635E run feedback available and Conditioning Timer has elapsed then go to step S0400
S0400	Drain E2635E	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Opening DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Provide Drain command to EM_EIE_PRODUCT1



Step	Description	Action	Action Description
		A0110	Waiting for LSL_E1235E-1 and LSLL_E1235E-1 to be in alarm
		A0120	Set Delay Time
		A0130	Start Delay Time
		A0140	Delay Time Message
		A0150	Set Task Pointer
T0500A	Blow PE2635E		If (LSL_E1235E-1 and LSLL_E1235E-1 in alarm and delay time 10 sec) or push button <Stop> (OP003) then go to step S0500
S0500	Blow PE2635E	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Waiting for EM_EIE_PRODUCT1 not to be in step Blow
		A0110	Set Task Pointer
T0600A	Ready		If PRODUCT1 is not in step <Blow> then go to step S0600
S0600	Ready	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4



Step	Description	Action	Action Description
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set Task Pointer
		A0110	Waiting for Startup in step Production or Drain and Reactor status
T0700A	Fill Oil		If (E2600 Startup is in step <Production> or <Drain>) and (previous E2603x is in step EIE) or (no E2603x in EIE and push button <Start> (OP002) then go to step S0700
T1100B	Close valves		If push button <Stop> (OP003) is selected then go to step S1000A
S0700	Fill Oil	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Waiting for LSH1103X1 to be in alarm
		A0110	Set Task Pointer
T0800A	EIE		If LSH1103X1 in alarm then go to step S0800
S0800	EIE	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening DEV2

Step	Description	Action	Action Description
		A0030	Enabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Set Task Pointer
T0900A	Blow		If first reactor is in series and push button <Blow> (OP005) is selected then go to step S0900
T0900A1	Close valves		If last reactor is in series and push button <Stop> (OP003) is selected then go to step S1000A
S0900	Blow	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0120	Set Blow Time
		A0130	Start Blow Time
		A0140	Delay Blow Message
		A0150	Set Task Pointer
T1000A	Clean		If LSL1103X is in alarm and Blow Timer has elapsed then go to step S1000
S1000	Clean	A0010	Closing and Disabling Conditional Logic for DEV1

Step	Description	Action	Action Description
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Reset the Reactor Sequence
		A0110	Waiting for Clean End or Clean Skip Button and Set Dynamic Reference
		A0120	Provide Aux Off command to Startup EM
		A0150	Set Task Pointer
T0600C	Ready		If push button <CLEAN END> (OP006) is selected then go to step S0600
T1100A	Close valves		If push button <CLEAN SKIP> (OP007) is selected then go to step S1000A
S1000A	Close valves	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9



Step	Description	Action	Action Description
		A0100	Reset Parameters
		A0110	Set Task Pointer
T1100A1	Disable HM and SM		If no pending actions in S1000A then go to step S1100
S1100	Disable HM and SM	A0010	Disable HM 01 (Module Ownership Lost)
		A0020	Reset Enzymes
		A0030	Set Task Pointer
T1200	Release Devices		If no pending actions in S1100 then go to step S1200
S1200	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
T9900	Complete		If no pending actions in S1200 then go to step S9900
S9900	Complete	A0010	Set Message: Blow Command Complete
		A0020	Self Release the EM & Reset First Pass
END	Command Complete		True

Table 20: Process Fluid Sequence

3.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Reactor Sequence	Floating Point	NA	0

OP002	Start	Named Set	NA	OFF
OP003	Stop	Named Set	NA	OFF
OP004	EIE	Named Set	NA	OFF
OP005	Blow	Named Set	NA	OFF
OP006	Clean End	Named Set	NA	OFF
OP007	Clean Skip	Named Set	NA	OFF
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Blow Timer	Floating Point	sec	5
TP003	Conditioning Timer	Floating Point	sec	5
TP004	Enzymes	Boolean	NA	False
TP005	Delay Time	Floating Point	sec	10

Table 21: Operating and Tuning Parameters

3.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference			
MY_TAG		EI1EQ_2603A	EI1EQ_2603B	EI1EQ_2603C	EI1EQ_2603D
OPR_CMD_MASK		7			
ESM	String	ESM_EIE_01			
DY_REF_1	Dynamic Reference				
DY_REF_2	Dynamic Reference				
DY_REF_3	Dynamic Reference				
DY_REF_4	Dynamic Reference				
DY_REF_5	Dynamic Reference				
STARTUP	String	EI1EQ_2600E			
PRODUCT1	String	EI1EQ_2635E			
LSH1103X1	External Reference	EI1_LSH1103A1/DI1/DISC_ACT			
LSL_E1235E-1	External Reference	EI1_LSL1235E1/DI1/DISC_ACT			
LSLL_E1235E-1	External Reference	EI1_LSLL1235E1/DI1/DISC_ACT			
LSL1103X	External Reference	EI1_LSL1103A/DI1/DISC_ACT			

Table 22: Configurable Parameters

3.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_EIE_REACTOR	Enzymatic Interesterification Area: Reactor

Table 23: Named Set Details

Details of states for NS_EIE_REACTOR named set are as follows.

Named Set Value	Named Set Name
00	Hold
01	Process Fluid

Table 24: Name Set State Details

4 EQUIPMENT MODULE CLASS DESIGN – EM_EIE_PRODUCT1 (E2635E)

4.1 Functional Description

EM_EIE_PRODUCT1 is a class based equipment module that adds enzyme to the tank and later executes deaeration and blow to complete the process. EM_EIE_PRODUCT1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Add Enzyme Command and should be released on completion of Drain Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

4.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_EIE_PRODUCT1	EI1EQ_2635E
DEV1	_V12_IP_130_SE	EI1_V1246AD
DEV2	_V12_IP_130_SE	EI1_V1210ED
DEV3	_V12_IP_130_SE	EI1_V1235ED
DEV4	_V12_IP_130_SE	EI1_V1235EF1
DEV5	_V12_IP_130_SE	EI1_V1235EF2
DEV6	_V12_IP_130_SE	EI1_V1235ER
DEV7	_V12_IP_130_SE	EI1_V1235EV
DEV8	_V12_IP_130_SE	EI1_V1235EW
DEV9	_V12_IP_130_SE	EI1_V1135EBN
DEV10	_V12_IP_130_SE	EI1_V1235ED1
DEV11	_V12_IP_130_SE	EI1_V1235ED2
DEV12	_MRSS11_130_SE	EI1_AG1235E
DEV13	_MRSS11_130_SE	EI1_PU1235E
DEV14	_MRSS11_130_SE	EI1_VP1241X
DEV15	_CSD_130_SE	EI1_PT1235E1

Table 25: Instances

4.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

4.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Stopping and Disabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Stopping DEV14
		A0150	Set DEV15 Mode to Remote OUT
		A0160	Set DEV15 output to Closed and Disable Conditional Logic



Step	Description	Action	Action Description
T9900A	Actions Complete or Time Out & HOLD=TRUE		Go to step H9900A
T9900B	Actions Complete or Time Out & HOLD=FALSE		Go to step H9900B
H9900A	Command Hold End	A0010	Set Message: Hold Command Complete
		A0020	Reset Hold Request
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
		A0140	Release Private Devices
		A0150	Release Private Devices
		A0160	Release Private Devices
END1	Command Complete		True

Table 26: Hold Sequence

4.3.2 Command 01 – Add Enzyme

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Add Enzyme Command.
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Acquire Private Devices
		A0120	Acquire Private Devices
		A0130	Acquire Private Devices
		A0140	Acquire Private Devices
		A0150	Acquire Private Devices
		A0160	Acquire Private Devices
		A0170	Reset Hold Request
T0100	Standby		Go to step S0100
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Fill Blow		If Task Pointer = 3 then go to step S0300
T0400B	Add Enzyme		If Task Pointer = 4 then go to step S0400
T0500B	Refill		If Task Pointer = 5 then go to step S0500
T0600B	Refill Blow		If Task Pointer = 6 then go to step S0600
T0700B	Deaeration		If Task Pointer = 7 then go to step S0700
T0800B	Ready		If Task Pointer = 8 then go to step S0800



Step	Description	Action	Action Description
T0900B	Conditioning		If Task Pointer = 9 then go to step S0900
S0100	Standby	A0010	Set DEV1 Mode to Cascade
		A0020	Set DEV2 Mode to Cascade
		A0030	Set DEV3 Mode to Cascade
		A0040	Set DEV4 Mode to Cascade
		A0050	Set DEV5 Mode to Cascade
		A0060	Set DEV6 Mode to Cascade
		A0070	Set DEV7 Mode to Cascade
		A0080	Set DEV8 Mode to Cascade
		A0090	Set DEV9 Mode to Cascade
		A0100	Set DEV10 Mode to Cascade
		A0110	Set DEV11 Mode to Cascade
		A0120	Set DEV12 Mode to Cascade
		A0130	Set DEV13 Mode to Cascade
		A0140	Set DEV14 Mode to Cascade
		A0150	Set DEV15 Mode to Remote Out
		A0160	Set DEV15 output to 0
		A0170	Closing and Disabling Conditional Logic for DEV1
		A0180	Closing and Disabling Conditional Logic for DEV2
		A0190	Closing DEV3
		A0200	Closing and Disabling Conditional Logic for DEV4
		A0210	Closing and Disabling Conditional Logic for DEV5
		A0220	Closing DEV6
		A0230	Enabling Conditional Logic for DEV7
		A0240	Closing and Disabling Conditional Logic for DEV8
		A0250	Closing and Disabling Conditional Logic for DEV9
		A0260	Closing and Disabling Conditional Logic for DEV10
		A0270	Closing and Disabling Conditional Logic for DEV11
		A0280	Stopping and Disabling Conditional Logic for DEV12



Step	Description	Action	Action Description
		A0290	Stopping and Disabling Conditional Logic for DEV13
		A0300	Stopping and Disabling Conditional Logic for DEV14
T0200A	Fill		If push button <Fill> (OP002) is selected then go to step S0200
S0200	Fill	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Starting DEV14
		A0150	Set DEV15 Mode to Remote Out
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Waiting LS_235E1 to not be in alarm
		A0180	Set Task Pointer
T0300A	Fill Blow		If LS_235E1 is not in alarm or push button <Stop> (OP002) is selected then go to step S0300
S0300	Fill Blow	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3



Step	Description	Action	Action Description
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Starting DEV14
		A0150	Set DEV15 Mode to Remote Out
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Set Blower Time
		A0180	Start Blower Time
		A0190	Blower Time Message
		A0200	Set Task Pointer
T0400A	Add Enzyme		If Fill Blow timer (TP002) has elapsed then go to step S0400
S0400	Add Enzyme	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8



Step	Description	Action	Action Description
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Starting DEV14
		A0150	Set DEV15 Mode to Remote Out
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Set Task Pointer
T0500A	Refill		If push button <Refill> (OP002) is selected then go to step S0500
S0500	Refill	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Starting DEV14
		A0150	Set DEV15 Mode to Remote Out

Step	Description	Action	Action Description
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Checking LSH_235E1 is in alarm
		A0180	Set Task Pointer
T0600A	Refill Blow		If LSH_235E1 in alarm or push button <Stop> (OP002) is selected then go to step S0600
S0600	Refill Blow		
		A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Starting DEV14
		A0150	Set DEV15 Mode to Remote Out
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Set Refill Blower Time
		A0180	Start Blower Time
		A0190	Blower Time Message
		A0200	Set Task Pointer
T0700A	Deaeration		If Fill Blow timer (TP002) has elapsed then go to step S0700
S0700	Deaeration	A0010	Enabling Conditional Logic for DEV1

Step	Description	Action	Action Description
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Starting DEV14
		A0150	Set DEV15 Mode to Remote Out
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Set Dearation Time
		A0180	Start Dearation Time
		A0190	Dearation Time Message
		A0200	Set Task Pointer
T0800A	Ready		If Deaeration timer (TP003) has elapsed then go to step S0800
S0800	Ready	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5



Step	Description	Action	Action Description
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Stopping DEV14
		A0150	Set DEV15 Mode to Remote Out
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Set Task Pointer
T0900A	Conditioning		If any E2603x is in step <Fill Enzyme> and no E2603x is in <Drain E2635E> then go to step S0900
S0900	Conditioning	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Opening DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10



Step	Description	Action	Action Description
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Stopping DEV14
		A0150	Set DEV15 Mode to Remote Cascade
		A0160	Set DEV15 setpoint to TP005_VALUE
		A0170	Set Task Pointer
T9900	Complete		If any E2603x is in <Drain E2635E> then go to step S9900
S9900	Complete	A0010	Dummy
END	Command Complete		True

Table 27: Add Enzyme Sequence

4.3.3

Command 02 – Drain

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Drain Command
T0100	Drain		If Task Pointer = 9 then go to step S0100
T0200B	Blow		If Task Pointer = 10 then go to step S0200
T0300B	Close Devices		If Task Pointer = 11 then go to step S0300
T0400B	Disable HM & SM		If Task Pointer = 12 then go to step S0400
S0100	Drain	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Opening DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Opening DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8

Step	Description	Action	Action Description
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Enabling Conditional Logic for DEV10
		A0110	Enabling Conditional Logic for DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Stopping DEV14
		A0150	Set DEV15 Mode to Remote Cascade
		A0160	Setting DEV1 setpoint to (Pressure RSP) and Enabling Conditional Logic
T0200A	Blow		If any E2603x is not in <Drain E2635E> then go to step S9900
S0200	Blow	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Opening DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Enabling Conditional Logic for DEV10
		A0110	Enabling Conditional Logic for DEV11
		A0120	Disabling Conditional Logic for DEV12
		A0130	Disabling Conditional Logic for DEV13
		A0140	Stopping DEV14
		A0150	Set DEV15 Mode to Remote Out
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Set Blower Time
		A0180	Start Blower Time
		A0190	Blower Time Message

Step	Description	Action	Action Description
		A0200	Set Task Pointer
T0300A	Close Devices		If Drain Blow Timer (TP004) has elapsed then go to step S0300
S0300	Close Devices	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Stopping and Disabling Conditional Logic for DEV12
		A0130	Stopping and Disabling Conditional Logic for DEV13
		A0140	Stopping DEV14
		A0150	Set DEV15 Mode to Remote OUT
		A0160	Set DEV15 output to Closed and Disable Conditional Logic
		A0170	Set Task Pointer
T0400A	Disable HM & SM		If no pending actions in S0300 then go to step S0400
S0400	Disable HM & SM	A0010	Disable HM 01 (Module Ownership Lost)
		A0020	Set Task Pointer
T0500	Release Devices		If no pending actions in S0400 then go to step S0500



Step	Description	Action	Action Description
S0500	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
		A0140	Release Private Devices
		A0150	Release Private Devices
		A0160	Release Private Devices
T9900	Complete		If no pending actions in S0500 then go to step S9900
S9900	Complete	A0010	Deaeration Command Complete
END	Command Complete		True

Table 28: Drain Sequence

4.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Fill Source	Named Set	N/A	Fill From E2682B
OP002	Command	Named Set	N/A	Stop
OP003	Drain Selection	Named Set	N/A	Drain to E2682B
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Fill Blow Timer	Floating Point	sec	50

Parameter	Description	Type	Units	Default Value/Reference
TP003	Deaeration Timer	Floating Point	min	10
TP004	Drain Blow Timer	Floating Point	sec	180

Table 29: Operating and Tuning Parameters

4.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		EI1EQ-2635E
OPR_CMD_MASK		7
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
ESM	String	ESM_EIE_01
LSL_235E1	External Reference	EI1_LSL1235E1/DI1/DISC_ACT
LSH_235E1	External Reference	EI1_LSH1235E1/DI1/DISC_ACT
LS_235E1	External Reference	EI1_LS1235E1/DI1/DISC_ACT
PT_235E1	External Reference	EI1_PT1235E1/PID1/HI_HI_ACT
STEP_TIME	Floating Point	
COMPLETED_TIME	Floating Point	
RESET_COMP_TM	Boolean	
HOLD	External Reference	EI1EQ_2600E/OP003_VALUE

Table 30: Configurable Parameters

4.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_EIE_PRODUCT1	Enzymatic Interesterification Area: Products
NS_EIE_FILL_SRC	Enzymatic Interesterification Area: Fill Source
NS_EIE_PUSH_BTN	Enzymatic Interesterification Area: Push Button

Table 31: Named Set Details

Details of states for NS_EIE_PRODUCT1 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Add Enzyme
02	Drain

Table 32: Name Set State Details

5 EQUIPMENT MODULE CLASS DESIGN – EM_EIE_PRODUCT2 (E2682B)

5.1 Functional Description

EM_EIE_PRODUCT2 is a class based equipment module that interacts with EM_EIE_REACTOR to perform Filling operation followed by Recirculation in order to complete Enzymatic Interesterification. EM_EIE_PRODUCT2 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Fill Command and should be released on completion of the same. Hold Request shall be enabled for all acquired devices at the end of Hold command.

5.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name	Instance Name
EM	EM_EIE_PRODUCT2	EI1EQ_2682B1	EI1EQ_2682B2
DEV1	_V12_IP_130_SE	EI1_V1216BFY	
DEV2	_V12_IP_130_SE	EI1_V1282B1F	EI1_V1282B2F
DEV3	_V12_IP_130_SE	EI1_V1282B1D	EI1_V1282B2D
DEV4	_V12_IP_130_SE	EI1_V1282B1R	EI1_V1282B2R
DEV5	_V12_IP_130_SE	EI1_V1282B1V	EI1_V1282B2V
DEV6	_V12_IP_130_SE	EI1_V1282B2BN	
DEV7	_V12_IP_130_SE	Ignored	
DEV8	_MRSS11_130_SE	EI1_PU1282B	

Table 33: Instances

5.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

5.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Common Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
T9900A	Actions Complete or Time Out & HOLD=TRUE		Go to step H9900A
T9900B	Actions Complete or Time Out & HOLD=FALSE		Go to step H9900B
H9900A	Command Hold End	A0010	Set Message: Hold Command Complete
		A0020	Reset Hold Request
END	Command Complete		True
H9900B		A0010	Release Devices Message

Step	Description	Action	Action Description
	Release Devices	A0020	Release Shared Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Shared Devices
		A0080	Release Shared Devices
		A0090	Release Shared Devices
END1	Command Complete		True

Table 34: Hold Sequence

5.3.2 Command 01 – Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Shared Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Shared Devices
		A0080	Acquire Shared Devices
		A0090	Acquire Shared Devices
T0100	Standby		Go to step S0100
S0100	Standby	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2

Step	Description	Action	Action Description
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
T0200A	Fill		If <Fill> latch (OP001) is ON then go to step S0200
T0300A	Recirculation		If <Recirculation> latch (OP003) is ON and <DRN_LINE_CLR> is ON then go to step S0300
T0400A	Fill E2635E		If <E2635E Fill Demand> is ON and <DRN_LINE_CLR> is ON then go to step S0400
T0500A	Drain		If <Drain> latch (OP004) is ON and <DRN_LINE_CLR> is ON then go to step S0500
S0200	Fill	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
T0300B	Recirculation		If <Recirculation> latch (OP003) is ON and <DRN_LINE_CLR> is ON then go to step S0300
T0500B	Drain		If <Drain> latch (OP004) is ON and <DRN_LINE_CLR> is ON then go to step S0500
T0100A	Standby		If <Drain> (OP004) and <Fill> latch (OP001) is OFF then go step S0100
T0400B	Fill E2635E		If <E2635E_FILL_DMND> is ON and <DRN_LINE_CLR> is ON then go to step S0400



Step	Description	Action	Action Description
S0300	Recirculation	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Opening and Disabling Conditional Logic for DEV3
		A0040	Opening DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
T0100B	Standby		If <Fill> latch (OP001) is OFF and (<Recirculation> latch (OP003) is OFF or <E2635E_FILL_DMND> is ON or <Drain> latch (OP004) is ON) then go to step S0100
T0200B	Fill		If <Fill> latch (OP001) is ON and (<Recirculation> latch (OP003) is OFF or <E2635E_FILL_DMND> is ON or <Drain> latch (OP004) is ON) then go to step S0200
S0400	Fill E2635E	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
T0100D	Standby		If <Fill> latch is OFF and <E2635E_FILL_DMND> is OFF then go to step S0100
T0200C	Fill		If <Fill> latch is ON and <E2635E_FILL_DMND> is OFF then go to step S0200
S0500	Drain	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Opening and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8



Step	Description	Action	Action Description
T0600A	Blow		If LSL1282B is in alarm or <Drain> latch (OP004) is OFF then go to step S0600
S0600	Blow	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
		A0090	Set Blower Time
		A0100	Start Blower Time
		A0110	Blower Time Message
T0200D	Fill		If <Fill> latch (OP001) is ON and Blow Timer has elapsed then go to step S0700
T0700	Standby		If <Fill> latch (OP001) is OFF and Blow Timer has elapsed then go to step S0200
S0700	Standby	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Stopping and Disabling Conditional Logic for DEV8
T0800	Release Devices		If no pending actions in S0700 then go to step S0800
S0800	Release Devices	A0010	Release Devices Message
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0100	Disable HM 01 (Module Ownership Lost)
T9900	Complete		If no pending actions S0800 then go to step S9900

Step	Description	Action	Action Description
S9900	Complete	A0010	Set Message: Fill Command Complete
		A0020	Self Release the EM & Reset First Pass
END	Command Complete		True

Table 35: Fill Sequence

5.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Available on Detail Display
Operating Parameters				
OP001	Fill	Named Set	N/A	OFF
OP002	NA	Floating Point	N/A	0
OP003	Recirculation	Named Set	N/A	OFF
OP004	Drain	Named Set	N/A	OFF
OP005	Fill E2635E	Named Set	N/A	OFF
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	0
TP002	Blow Time	Floating Point	sec	0

Table 36: Operating and Tuning Parameters

5.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference	
MY_TAG		EI1EQ_2682B1	EI1EQ_2682B2
OPR_CMD_MASK		3	
HOLD	External Reference	EI1EQ_2600E/OP003_VALUE	
LSL1282B	External Reference	EI1_LSL1282B/DI1/DISC_ACT	
LT1282BX1	External Reference	EI1_LT1282B11/AI1/HI_HI_ACT	EI1_LT1282B21/AI1/HI_HI_ACT

Parameter Name	Parameter Type	Value/Reference	
LSH1282BX1	External Reference	EI1_LSH1282B11/DI1/DISC_ACT	EI1_LSH1282B21/DI1/DISC_ACT
E2635E_FILL_DMND	External Reference	EI1EQ_2635E/FILL_DEMAND/OUT_D	EI1EQ_2635E/FILL_DEMAND/OUT_D
DRN_LINE_CLR	External Reference	ESM_EIE_01/DRN_LINE_CLR/OUT_D	ESM_EIE_01/DRN_LINE_CLR/OUT_D
STEP_TIME	Floating Point		
COMPLETED_TIME	Floating Point		
RESET_COMP_TM	Boolean		
LSH1282BX	External Reference		

Table 37: Configurable Parameters

5.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_EIE_PRODUCT2	Enzymatic Interesterification Area: Product2

Table 38: Named Set Details

Details of states for NS_EIE_PRODUCT2 named set are as follows.

Named Set Value	Named Set Name
00	Hold
01	Fill

Table 39: Name Set State Details

6 MODULE FORCE SETPOINT

6.1 Functional Description

Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_AG1235E	Condition 1	Conditional Logic	('TP01.CV' = TRUE) AND '//EI1_LSL1235E1/DI1/DISC_ACT.CV' = FALSE	START	FALSE	0
EI1_AG1235E	Condition 2	Conditional Logic	('TP01.CV' = TRUE) AND '//EI1_LSL1235E1/DI1/DISC_ACT.CV' = TRUE	STOP	FALSE	0
EI1_PU1034NA	Condition 1	Conditional Logic	'TP01.CV' = TRUE AND '//EI1EQ_2682A/OP001_VALUE.CV' = '_DO_OFF_ON:ON' AND '//MAJOR_ALM/FAULT_NO.CV' = 0	START	FALSE	0
EI1_PU1034NA	Condition 2	Conditional Logic	'TP01.CV' = TRUE AND ('//EI1EQ_2682A/OP001_VALUE.CV' = '_DO_OFF_ON:OFF' OR '//MAJOR_ALM/FAULT_NO.CV' != 0)	STOP	FALSE	0
EI1_PU105081A	Condition 1	Conditional Logic	('TP01.CV' = TRUE) AND '//EI1_PU1082A/PV_D.CV' = '_EDC_MTR2_PV:RUNNING'	START	FALSE	0
EI1_PU105081A	Condition 2	Conditional Logic	('TP01.CV' = TRUE) AND '//EI1_PU1082A/PV_D.CV' != '_EDC_MTR2_PV:RUNNING'	STOP	FALSE	0
EI1_PU1082A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND '//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND '//EI1_LSL1082A1/DI1/DISC_ACT.CV' != TRUE AND '//MAJOR_ALM/FAULT_NO.CV' = 0	START	FALSE	0
EI1_PU1082A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' != FALSE OR '//EI1_LSL1082A1/DI1/DISC_ACT.CV' = TRUE OR '//MAJOR_ALM/FAULT_NO.CV' != 0)	STOP	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_PU115003	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1_LSL1150/DI1/DISC_ACT.CV' != TRUE AND //EI1EQ_2600E/OP004_VALUE.CV' = 'NS_ENABLE:ENABLE')	START	FALSE	5
EI1_PU115003	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1_LSL1150/DI1/DISC_ACT.CV' = TRUE)	STOP	FALSE	120
EI1_PU115003	Condition 3	Conditional Logic	(/TP01.CV' = TRUE) AND //EI1EQ_2600E/OP004_VALUE.CV' = 'NS_ENABLE:DISABLE'	STOP	FALSE	0
EI1_PU1235E	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND (//EI1_V1103AR/PV_D.CV' = '_VLVNC_PV:OPEN' AND //EI1_LSL1103A/DI1/DISC_ACT.CV' = FALSE) OR (//EI1_V1103BR/PV_D.CV' = '_VLVNC_PV:OPEN' AND //EI1_LSL1103B/DI1/DISC_ACT.CV' = FALSE) OR (//EI1_V1103CR/PV_D.CV' = '_VLVNC_PV:OPEN' AND //EI1_LSL1103C/DI1/DISC_ACT.CV' = FALSE)OR (//EI1_V1103DR/PV_D.CV' = '_VLVNC_PV:OPEN' AND //EI1_LSL1103D/DI1/DISC_ACT.CV' = FALSE)))	START	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_PU1235E	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (//EI1_V1103AR/PV_D.CV' = '_VLVNC_PV:OPEN' AND //EI1_LSL1103A/DI1/DISC_ACT.CV' = TRUE) OR (//EI1_V1103BR/PV_D.CV' = '_VLVNC_PV:OPEN' AND //EI1_LSL1103B/DI1/DISC_ACT.CV' = TRUE) OR (//EI1_V1103CR/PV_D.CV' = '_VLVNC_PV:OPEN' AND //EI1_LSL1103C/DI1/DISC_ACT.CV' = TRUE)OR (//EI1_V1103DR/PV_D.CV' = '_VLVNC_PV:OPEN' AND //EI1_LSL1103D/DI1/DISC_ACT.CV' = TRUE) OR (//EI1_V1103AR/PV_D.CV' != '_VLVNC_PV:OPEN' AND //EI1_V1103BR/PV_D.CV' != '_VLVNC_PV:OPEN' AND //EI1_V1103CR/PV_D.CV' != '_VLVNC_PV:OPEN' AND //EI1_V1103DR/PV_D.CV' != '_VLVNC_PV:OPEN'))	STOP	FALSE	0
EI1_PU1235E	Condition 3	Conditional Logic	(/TP02.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND (//EI1EQ_2603A/COMMAND_00001/S0400/ACTIVE.CV' = FALSE OR //EI1_LSL1103A/DI1/DISC_ACT.CV' = FALSE) AND (//EI1EQ_2603B/COMMAND_00001/S0400/ACTIVE.CV' = FALSE OR //EI1_LSL1103B/DI1/DISC_ACT.CV' = FALSE) AND (//EI1EQ_2603C/COMMAND_00001/S0400/ACTIVE.CV' = FALSE OR //EI1_LSL1103C/DI1/DISC_ACT.CV' = FALSE) AND (//EI1EQ_2603D/COMMAND_00001/S0400/ACTIVE.CV' = FALSE OR //EI1_LSL1103D/DI1/DISC_ACT.CV' = FALSE)) AND (//EI1_V1235ED1/PV_D.CV' = '_VLVNC_PV:OPEN' OR //EI1_V1235ED2/PV_D.CV' = '_VLVNC_PV:OPEN'))	START	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_PU1235E	Condition 4	Conditional Logic	(/TP02.CV' = TRUE) AND '//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (//EI1EQ_2603A/COMMAND_00001/S0400/ACTIVE.CV' = TRUE AND '//EI1_LSL1103A/DI1/DISC_ACT.CV' = TRUE) OR (//EI1EQ_2603B/COMMAND_00001/S0400/ACTIVE.CV' = TRUE AND '//EI1_LSL1103B/DI1/DISC_ACT.CV' = TRUE) OR (//EI1EQ_2603C/COMMAND_00001/S0400/ACTIVE.CV' = TRUE AND '//EI1_LSL1103C/DI1/DISC_ACT.CV' = TRUE) OR (//EI1EQ_2603D/COMMAND_00001/S0400/ACTIVE.CV' = TRUE AND '//EI1_LSL1103D/DI1/DISC_ACT.CV' = TRUE) OR (//EI1_V1235ED1/PV_D.CV' != '_VLVNC_PV:OPEN' AND '//EI1_V1235ED2/PV_D.CV' != '_VLVNC_PV:OPEN')	STOP	FALSE	0
EI1_PU1282B	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1_LSL1282B/DI1/DISC_ACT' != TRUE) AND '//EI1EQ_2600E/OP003_VALUE.CV' = '_DO_OFF_ON:OFF' AND (//EI1_V1282B1D/PV_D.CV' = '_VLVNC_PV:OPEN' OR '//EI1_V1282B2D/PV_D.CV' = '_VLVNC_PV:OPEN')	START	FALSE	2
EI1_PU1282B	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1_LSL1282B/DI1/DISC_ACT' = TRUE OR '//EI1EQ_2600E/OP003_VALUE.CV' = '_DO_OFF_ON:ON' OR (//EI1_V1282B1D/PV_D.CV' != '_VLVNC_PV:OPEN' AND '//EI1_V1282B2D/PV_D.CV' != '_VLVNC_PV:OPEN'))	STOP	FALSE	0
EI1_PU1282B	Condition 3	Conditional Logic	(/TP02.CV' = TRUE) AND (//EI1_LSL1282B/DI1/DISC_ACT' != TRUE) AND '//EI1EQ_2600E/OP003_VALUE.CV' = '_DO_OFF_ON:OFF' AND '//EI1_V1282B1D/PV_D.CV' = '_VLVNC_PV:OPEN' AND '//EI1_V1235EF1/PV_D.CV' = '_VLVNC_PV:OPEN' AND (//EI1EQ_2635E/COMMAND_00001/S0200/ACTIVE.CV' OR '//EI1EQ_2635E/COMMAND_00001/S0500/ACTIVE.CV')	START	FALSE	0



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EI1_PU1282B	Condition 4	Conditional Logic	(('/TP02.CV' = TRUE) AND ('//EI1_LSL1282B/DI1/DISC_ACT' = TRUE OR '//EI1EQ_2600E/OP003_VALUE.CV' = '_DO_OFF_ON:ON' OR '//EI1_V1282B1D/PV_D.CV' != '_VLVNC_PV:OPEN' OR '//EI1_V1235EF1/PV_D.CV' != '_VLVNC_PV:OPEN' OR ('//EI1EQ_2635E/COMMAND_00001/S0200/ACTIVE.CV' = FALSE AND '//EI1EQ_2635E/COMMAND_00001/S0500/ACTIVE.CV' = FALSE))	STOP	FALSE	0
EI1_PU1282B	Condition 5	Conditional Logic	(('/TP03.CV' = TRUE) AND ('//EI1EQ_2682B1/OWNER_ID.CV' = ""EI1EQ_2600E"" AND '//EI1EQ_2682B2/COMMAND_00001/S0300/ACTIVE.CV' = TRUE OR '//EI1EQ_2682B2/COMMAND_00001/S0400/ACTIVE.CV' = TRUE OR '//EI1EQ_2682B2/COMMAND_00001/S0500/ACTIVE.CV' = TRUE) OR ('//EI1EQ_2682B2/OWNER_ID.CV' = ""EI1EQ_2600E"" AND '//EI1EQ_2682B1/COMMAND_00001/S0300/ACTIVE.CV' = TRUE OR '//EI1EQ_2682B1/COMMAND_00001/S0400/ACTIVE.CV' = TRUE OR '//EI1EQ_2682B1/COMMAND_00001/S0500/ACTIVE.CV' = TRUE)	START	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_PU1282B	Condition 6	Conditional Logic	(/TP03.CV' = TRUE) AND (//EI1EQ_2682B1/OWNER_ID.CV' = ""EI1EQ_2600E"" AND //EI1EQ_2682B2/COMMAND_00001/S0300/ACTIVE.CV' != TRUE AND //EI1EQ_2682B2/COMMAND_00001/S0400/ACTIVE.CV' != TRUE AND //EI1EQ_2682B2/COMMAND_00001/S0500/ACTIVE.CV' != TRUE) OR (//EI1EQ_2682B2/OWNER_ID.CV' = ""EI1EQ_2600E"" AND //EI1EQ_2682B1/COMMAND_00001/S0300/ACTIVE.CV' != TRUE AND //EI1EQ_2682B1/COMMAND_00001/S0400/ACTIVE.CV' != TRUE AND //EI1EQ_2682B1/COMMAND_00001/S0500/ACTIVE.CV' != TRUE)	STOP	FALSE	0
EI1_V1082A	Condition 1	Conditional Logic	/TP01.CV' = TRUE AND //EI1_V1082AD/PV_D.CV' = ' _EDC_VLV_PV:CLOSED' AND //ESM_EIE_01/MAJOR_ALM/OUT_D.CV' = FALSE AND /PV_D.CV' = ' _EDC_VLV_PV:CLOSED'	OPEN	FALSE	5
EI1_V1082A	Condition 2	Conditional Logic	/TP01.CV' = TRUE AND (//EI1_V1082AD/PV_D.CV' != ' _EDC_VLV_PV:CLOSED' OR //ESM_EIE_01/MAJOR_ALM/OUT_D.CV' != FALSE)	CLOSE	FALSE	0
EI1_V1082A	Condition 3	Conditional Logic	/TP01.CV' = TRUE AND //EI1_V1082AD/PV_D.CV' = ' _EDC_VLV_PV:CLOSED' AND //ESM_EIE_01/MAJOR_ALM/OUT_D.CV' = FALSE AND /PV_D.CV' = ' _EDC_VLV_PV:OPEN'	CLOSE	FALSE	5



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1082AF	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND '//EI1_LT1082A1/AI1/HI_HI_ACT.CV' != TRUE AND '//EI1_LSH1082A1/DI1/DISC_ACT.CV' != TRUE) AND '//EI1_V1235EF2/PV_D.CV' = ' EDC_VLV_PV:CLOSED'	OPEN	FALSE	0
EI1_V1082AF	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' != FALSE OR '//EI1_LT1082A1/AI1/HI_HI_ACT.CV' = TRUE OR '//EI1_LSH1082A1/DI1/DISC_ACT.CV' = TRUE) OR '//EI1_V1235EF2/PV_D.CV' != ' EDC_VLV_PV:CLOSED'	CLOSE	FALSE	0
EI1_V1082AR	Condition 1	Conditional Logic	'TP01.CV' = TRUE AND '//EI1EQ_2682A/OP002_VALUE.CV' = ' DO_OFF_ON:ON'	OPEN	FALSE	0
EI1_V1082AR	Condition 2	Conditional Logic	'TP01.CV' = TRUE AND '//EI1EQ_2682A/OP002_VALUE.CV' = '_DO_OFF_ON:OFF' AND ((//EI1EQ_2603A/OP001_VALUE.CV' != 1 OR '//EI1_V1103AF1/PV_D.CV' = ' _VLVNC_PV:OPEN') AND (//EI1EQ_2603B/OP001_VALUE.CV' != 1 OR '//EI1_V1103BF1/PV_D.CV' = ' _VLVNC_PV:OPEN') AND (//EI1EQ_2603C/OP001_VALUE.CV' != 1 OR '//EI1_V1103CF1/PV_D.CV' = ' _VLVNC_PV:OPEN') AND (//EI1EQ_2603D/OP001_VALUE.CV' != 1 OR '//EI1_V1103DF1/PV_D.CV' = ' _VLVNC_PV:OPEN'))	CLOSE	FALSE	0
EI1_V1082AV	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND '//EI1_PT1082A1/AI1/HI_HI_ACT.CV' = TRUE;	OPEN	FALSE	0
EI1_V1082AV	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND '//EI1_PT1082A1/AI1/HI_HI_ACT.CV' != TRUE;	CLOSE	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103ABN	Condition 1	Auto Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND (('//EI1EQ_2682B1/A_TARGET.CV' ='NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B1/A_STATE.CV' = '\$phase_state:Running') OR ('//EI1EQ_2682B2/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B2/A_STATE.CV' = '\$phase_state:Running')) AND ('/PV_D.CV' = ' _VLVNC_PV:CLOSED'))	OPEN	FALSE	5
EI1_V1103ABN	Condition 2	Auto Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR ('//EI1EQ_2682B1/A_TARGET.CV' != 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B1/A_STATE.CV' != '\$phase_state:Running' AND '//EI1EQ_2682B2/A_TARGET.CV' != 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B2/A_STATE.CV' != '\$phase_state:Running'))	CLOSE	FALSE	0
EI1_V1103ABN	Condition 3	Auto Logic	(('/TP01.CV' = TRUE) AND ('/PV_D.CV' = ' _VLVNC_PV:OPEN'))	CLOSE	FALSE	5

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103AD2	Condition 1	Auto Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND (('//EI1EQ_2603A/OP001_VALUE.CV' = 1 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 1) OR ('//EI1EQ_2603A/OP001_VALUE.CV' = 2 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 3) OR ('//EI1EQ_2603A/OP001_VALUE.CV' = 4 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 10) OR ('//EI1EQ_2603A/OP001_VALUE.CV' = 3 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 6)) AND (('//EI1EQ_2682B1/A_TARGET.CV' ='NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B1/A_STATE.CV' = '\$phase_state:Running') OR ('//EI1EQ_2682B2/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B2/A_STATE.CV' = '\$phase_state:Running'))))	OPEN	FALSE	0
EI1_V1103AD2	Condition 2	Auto Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (('//EI1EQ_2603A/OP001_VALUE.CV' != 1 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 1) OR ('//EI1EQ_2603A/OP001_VALUE.CV' != 2 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 3) OR ('//EI1EQ_2603A/OP001_VALUE.CV' != 4 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 10) OR ('//EI1EQ_2603A/OP001_VALUE.CV' != 3 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 6)) OR ('//EI1EQ_2682B1/ACT_STEPS.CV' = "" AND ('//EI1EQ_2682B2/ACT_STEPS.CV' = "")))	CLOSE	FALSE	0
EI1_V1103AF1	Condition 1	Auto Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2603A/OP001_VALUE.CV' = 1) AND '//EI1EQ_2682A/OP002_VALUE.CV' = ' DO_OFF_ON:OFF'	OPEN	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103AF1	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND (('/EI1EQ_2603A/OP001_VALUE.CV' != 1) OR ('/EI1EQ_2682A/OP002_VALUE.CV' = '_DO_OFF_ON:ON' AND '/EI1_V1082AR/PV_D.CV' = '_VLVNC_PV:OPEN'))	CLOSE	FALSE	0
EI1_V1103AF2	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND ('/EI1EQ_2603A/OP001_VALUE.CV' != 1) REM AND ('/EI1EQ_2603D/OP001_VALUE.CV' != 0)	OPEN	FALSE	0
EI1_V1103AF2	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND ' /EI1EQ_2603A/OP001_VALUE.CV' = 1 REM OR ' /EI1EQ_2603D/OP001_VALUE.CV' = 0)	CLOSE	FALSE	0
EI1_V1103AQE	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND ('/EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S0900"" OR ' /EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S9900"" OR ' /EI1EQ_2635E/A_PV.CV' = 'NS_EIE_PRODUCT1:Add Enzyme' OR ' /EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0000"" OR ' /EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0100"")	OPEN	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103AQE	Condition 2	Auto Logic	('/TP01.CV' = TRUE)AND ('/EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' != ""S0900"" AND '//EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' != ""S9900"" AND '//EI1EQ_2635E/A_PV.CV' != 'NS_EIE_PRODUCT1:Add Enzyme' AND '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' != ""S0000"" AND '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' != ""S0100"")	CLOSE	FALSE	0
EI1_V1103AV	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND ('/EI1_LSH1103A1/DI1/DISC_ACT' != TRUE)	OPEN	FALSE	0
EI1_V1103AV	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND ('/EI1_LSH1103A1/DI1/DISC_ACT' = TRUE)	CLOSE	FALSE	0
EI1_V1103BBN	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND ('/EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND ('/EI1EQ_2682B1/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B1/A_STATE.CV' = '\$phase_state:Running' OR '//EI1EQ_2682B2/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B2/A_STATE.CV' = '\$phase_state:Running') AND ('/PV_D.CV' = ' _VLVNC_PV:CLOSED'))	OPEN	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103BBN	Condition 2	Auto Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (//EI1EQ_2682B1/A_TARGET.CV' !='NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B1/A_STATE.CV' != '\$phase_state:Running' AND '//EI1EQ_2682B2/A_TARGET.CV' !='NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B2/A_STATE.CV' != '\$phase_state:Running'))	CLOSE	FALSE	0
EI1_V1103BBN	Condition 3	Auto Logic	(/TP01.CV' = TRUE) AND (/PV_D.CV' = ' _VLVNC_PV:OPEN')	OPEN	FALSE	0
EI1_V1103BD2	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND ((//EI1EQ_2603B/OP001_VALUE.CV' = 1 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 1) OR (//EI1EQ_2603B/OP001_VALUE.CV' = 2 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 3) OR (//EI1EQ_2603B/OP001_VALUE.CV' = 4 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 10) OR (//EI1EQ_2603B/OP001_VALUE.CV' = 3 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 6)) AND (//EI1EQ_2682B1/A_TARGET.CV' ='NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B1/A_STATE.CV' = '\$phase_state:Running') OR (//EI1EQ_2682B2/A_TARGET.CV' ='NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B2/A_STATE.CV' = '\$phase_state:Running'))))	OPEN	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103BD2	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (('//EI1EQ_2603B/OP001_VALUE.CV' != 1 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 1) OR ('//EI1EQ_2603B/OP001_VALUE.CV' != 2 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 3) OR ('//EI1EQ_2603B/OP001_VALUE.CV' != 4 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 10) OR ('//EI1EQ_2603B/OP001_VALUE.CV' != 3 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 6)) OR ('//EI1EQ_2682B1/ACT_STEPS.CV' = ' ' AND ('//EI1EQ_2682B2/ACT_STEPS.CV' = ' ')))	CLOSE	FALSE	0
EI1_V1103BF1	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1EQ_2603B/OP001_VALUE.CV' = 1) AND '//EI1EQ_2682A/OP002_VALUE.CV' = ' DO_OFF_ON:OFF'	OPEN	FALSE	0
EI1_V1103BF1	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND (('//EI1EQ_2603B/OP001_VALUE.CV' != 1) OR ('//EI1EQ_2682A/OP002_VALUE.CV' = ' DO_OFF_ON:ON' AND '//EI1_V1082AR/PV_D.CV' = ' VLVNC_PV:OPEN'))	CLOSE	FALSE	0
EI1_V1103BF2	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1EQ_2603B/OP001_VALUE.CV' != 1) AND ('//EI1EQ_2603A/OP001_VALUE.CV' != 0)	OPEN	FALSE	0
EI1_V1103BF2	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1EQ_2603B/OP001_VALUE.CV' = 1 OR '//EI1EQ_2603A/OP001_VALUE.CV' = 0)	CLOSE	FALSE	0
EI1_V1103BFY	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1EQ_2603B/OP001_VALUE.CV' != 1) AND ('//EI1EQ_2603A/OP001_VALUE.CV' = 0)	OPEN	FALSE	0
EI1_V1103BFY	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1EQ_2603B/OP001_VALUE.CV' != 1 OR '//EI1EQ_2603A/OP001_VALUE.CV' = 0)	CLOSE	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103BQE	Condition 1	Conditional Logic	('/TP01.CV' = TRUE)AND ('/EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S0900"" OR '//EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S9900"" OR '//EI1EQ_2635E/A_PV.CV' = 'NS_EIE_PRODUCT1:Add Enzyme' OR '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0000"" OR '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0100"")	OPEN	FALSE	0
EI1_V1103BQE	Condition 2	Conditional Logic	('/TP01.CV' = TRUE)AND ('/EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' != ""S0900"" AND '//EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' != ""S9900"" AND '//EI1EQ_2635E/A_PV.CV' != 'NS_EIE_PRODUCT1:Add Enzyme' AND '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' != ""S0000"" AND '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' != ""S0100"")	CLOSE	FALSE	0
EI1_V1103BV	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND ('/EI1_LSH1103B1/DI1/DISC_ACT' != TRUE)	OPEN	FALSE	0
EI1_V1103BV	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND ('/EI1_LSH1103B1/DI1/DISC_ACT' = TRUE)	CLOSE	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103CBN	Condition 1	Conditional Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND ('//EI1EQ_2682B1/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B1/A_STATE.CV' = '\$phase_state:Running' OR '//EI1EQ_2682B2/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B2/A_STATE.CV' = '\$phase_state:Running') AND ('/PV_D.CV' = ' _VLVNC_PV:CLOSED'))	CLOSE	FALSE	0
EI1_V1103CBN	Condition 2	Conditional Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR ('//EI1EQ_2682B1/A_TARGET.CV' != 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B1/A_STATE.CV' != '\$phase_state:Running' AND '//EI1EQ_2682B2/A_TARGET.CV' != 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B2/A_STATE.CV' != '\$phase_state:Running'))	OPEN	FALSE	0
EI1_V1103CBN	Condition 3	Conditional Logic	(('/TP01.CV' = TRUE) AND ('/PV_D.CV' = ' _VLVNC_PV:OPEN'))	CLOSE	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103CD2	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND (//EI1EQ_2603C/OP001_VALUE.CV' = 1 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 1) OR (//EI1EQ_2603C/OP001_VALUE.CV' = 2 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 3) OR (//EI1EQ_2603C/OP001_VALUE.CV' = 4 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 10) OR (//EI1EQ_2603C/OP001_VALUE.CV' = 3 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 6)) AND (//EI1EQ_2682B1/A_TARGET.CV' ='NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B1/A_STATE.CV' = '\$phase_state:Running') OR (//EI1EQ_2682B2/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B2/A_STATE.CV' = '\$phase_state:Running'))	OPEN	FALSE	0
EI1_V1103CD2	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (//EI1EQ_2603C/OP001_VALUE.CV' != 1 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 1) OR (//EI1EQ_2603C/OP001_VALUE.CV' != 2 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 3) OR (//EI1EQ_2603C/OP001_VALUE.CV' != 4 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 10) OR (//EI1EQ_2603C/OP001_VALUE.CV' != 3 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 6)) OR (//EI1EQ_2682B1/ACT_STEPS.CV' = "" AND (//EI1EQ_2682B2/ACT_STEPS.CV' = "")))	CLOSE	FALSE	0
EI1_V1103CF1	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2603C/OP001_VALUE.CV' = 1) AND '//EI1EQ_2682A/OP002_VALUE.CV' = ' DO_OFF_ON:OFF'	OPEN	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103CF1	Condition 2	Conditional Logic	('TP01.CV' = TRUE) AND (('EI1EQ_2603C/OP001_VALUE.CV' != 1) OR ('EI1EQ_2682A/OP002_VALUE.CV' = '_DO_OFF_ON:ON' AND 'EI1_V1082AR/PV_D.CV' = '_VLVNC_PV:OPEN'))	CLOSE	FALSE	0
EI1_V1103CF2	Condition 1	Conditional Logic	('TP01.CV' = TRUE) AND ('EI1EQ_2603C/OP001_VALUE.CV' != 1) AND ('EI1EQ_2603A/OP001_VALUE.CV' != 0 OR 'EI1EQ_2603B/OP001_VALUE.CV' != 0)	OPEN	FALSE	0
EI1_V1103CF2	Condition 2	Conditional Logic	('TP01.CV' = TRUE) AND ('EI1EQ_2603C/OP001_VALUE.CV' = 1 OR ('EI1EQ_2603A/OP001_VALUE.CV' = 0 AND 'EI1EQ_2603B/OP001_VALUE.CV' = 0))	CLOSE	FALSE	0
EI1_V1103CFY	Condition 1	Conditional Logic	('TP01.CV' = TRUE) AND ('EI1EQ_2603C/OP001_VALUE.CV' != 1) AND ('EI1EQ_2603B/OP001_VALUE.CV' = 0) AND ('EI1EQ_2603A/OP001_VALUE.CV' = 0)	OPEN	FALSE	0
EI1_V1103CFY	Condition 2	Conditional Logic	('TP01.CV' = TRUE) AND ('EI1EQ_2603C/OP001_VALUE.CV' = 1 OR 'EI1EQ_2603B/OP001_VALUE.CV' != 0 OR 'EI1EQ_2603A/OP001_VALUE.CV' != 0)	CLOSE	FALSE	0
EI1_V1103CQE	Condition 1	Conditional Logic	('TP01.CV' = TRUE) AND ('EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S0900"" OR 'EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S9900"" OR 'EI1EQ_2635E/A_PV.CV' = 'NS_EIE_PRODUCT1:Add Enzyme' OR 'EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0000"" OR 'EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0100""))	OPEN	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103CQE	Condition 2	Conditional Logic	('/TP01.CV' = TRUE)AND ('/EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' != ""S0900"" AND '//EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' != ""S9900"" AND '//EI1EQ_2635E/A_PV.CV' != 'NS_EIE_PRODUCT1:Add Enzyme' AND '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' != ""S0000"" AND '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' != ""S0100"")	CLOSE	FALSE	0
EI1_V1103CV	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND ('/EI1_LSH1103C1/DI1/DISC_ACT' != TRUE)	OPEN	FALSE	0
EI1_V1103CV	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND ('/EI1_LSH1103C1/DI1/DISC_ACT' = TRUE)	CLOSE	FALSE	0
EI1_V1103DBN	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND ('/EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND ('/EI1EQ_2682B1/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B1/A_STATE.CV' = '\$phase_state:Running' OR '//EI1EQ_2682B2/A_TARGET.CV' = 'NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B2/A_STATE.CV' = '\$phase_state:Running') AND ('/PV_D.CV' = ' _VLVNC_PV:CLOSED'))	OPEN	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103DBN	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (//EI1EQ_2682B1/A_TARGET.CV' !='NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B1/A_STATE.CV' != '\$phase_state:Running' AND '//EI1EQ_2682B2/A_TARGET.CV' !='NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B2/A_STATE.CV' != '\$phase_state:Running'))	CLOSE	FALSE	0
EI1_V1103DBN	Condition 3	Conditional Logic	(/TP01.CV' = TRUE) AND (/PV_D.CV' = ' _VLVNC_PV:OPEN')	OPEN	FALSE	0
EI1_V1103DD2	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND ((//EI1EQ_2603D/OP001_VALUE.CV' = 1 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 1) OR '//EI1EQ_2603D/OP001_VALUE.CV' = 2 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 3) OR '//EI1EQ_2603D/OP001_VALUE.CV' = 4 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 10) OR '//EI1EQ_2603D/OP001_VALUE.CV' = 3 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 6)) AND (('/EI1EQ_2682B1/A_TARGET.CV' ='NS_EIE_PRODUCT2:Fill' AND '//EI1EQ_2682B1/A_STATE.CV' = '\$phase_state:Running') OR (//EI1EQ_2682B2/A_TARGET.CV' ='NS_EIE_PRODUCT2:Fill' AND '/EI1EQ_2682B2/A_STATE.CV' = '\$phase_state:Running'))))	OPEN	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1103DD2	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (//EI1EQ_2603D/OP001_VALUE.CV' != 1 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 1) OR (//EI1EQ_2603D/OP001_VALUE.CV' != 2 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 3) OR (//EI1EQ_2603D/OP001_VALUE.CV' != 4 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 10) OR (//EI1EQ_2603D/OP001_VALUE.CV' != 3 AND '//ESM_EIE_01/REACT_SEQ_SUM.CV' = 6)) OR (//EI1EQ_2682B1/ACT_STEPS.CV' = "" AND (//EI1EQ_2682B2/ACT_STEPS.CV' = "")))	CLOSE	FALSE	0
EI1_V1103DF1	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2603D/OP001_VALUE.CV' = 1) AND '//EI1EQ_2682A/OP002_VALUE.CV' = ' DO_OFF_ON:OFF'	OPEN	FALSE	0
EI1_V1103DF1	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2603D/OP001_VALUE.CV' != 1) OR (//EI1EQ_2682A/OP002_VALUE.CV' = ' DO_OFF_ON:ON' AND '//EI1_V1082AR/PV_D.CV' = ' VLVNC_PV:OPEN'))	CLOSE	FALSE	0
EI1_V1103DF2	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND (//EI1EQ_2603D/OP001_VALUE.CV' != 1) REM AND (//EI1EQ_2603C/OP001_VALUE.CV' != 0)	OPEN	FALSE	0
EI1_V1103DF2	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND '//EI1EQ_2603D/OP001_VALUE.CV' = 1 REM OR '//EI1EQ_2603C/OP001_VALUE.CV' = 0)	CLOSE	FALSE	0



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EI1_V1103DQE	Condition 1	Conditional Logic	('/TP01.CV' = TRUE)AND ('/EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S0900"" OR '//EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S9900"" OR '//EI1EQ_2635E/A_PV.CV' = 'NS_EIE_PRODUCT1:Add Enzyme' OR '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0000"" OR '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0100"")	OPEN	FALSE	0
EI1_V1103DQE	Condition 2	Conditional Logic	('/TP01.CV' = TRUE)AND ('/EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S0900"" OR '//EI1EQ_2635E/COMMAND_00001/ACT_STEPS.CV' = ""S9900"" OR '//EI1EQ_2635E/A_PV.CV' = 'NS_EIE_PRODUCT1:Add Enzyme' OR '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0000"" OR '//EI1EQ_2635E/COMMAND_00002/ACT_STEPS.CV' = ""S0100"")	CLOSE	FALSE	0
EI1_V1103DV	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND ('/EI1_LSH1103D1/DI1/DISC_ACT' != TRUE)	OPEN	FALSE	0
EI1_V1103DV	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND ('/EI1_LSH1103D1/DI1/DISC_ACT' = TRUE)	CLOSE	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1135EBN	Condition 1	Conditional Logic	(/TP01.CV' = TRUE) AND '//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND (//EI1_V1235ED1/PV_D.CV' = '_VLVNC_PV:OPEN' OR '//EI1_V1235ED2/PV_D.CV' = '_VLVNC_PV:OPEN') AND '/PV_D.CV' = '_VLVNC_PV:CLOSED' AND (//EI1_V1282B1F/PV_D.CV' = '_VLVNC_PV:OPEN' OR '//EI1_V1282B2F/PV_D.CV' = '_VLVNC_PV:OPEN')	OPEN	FALSE	5
EI1_V1135EBN	Condition 2	Conditional Logic	(/TP01.CV' = TRUE) AND '//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR (//EI1_V1235ED1/PV_D.CV' != '_VLVNC_PV:OPEN' AND '//EI1_V1235ED2/PV_D.CV' != '_VLVNC_PV:OPEN') OR (//EI1_V1282B1F/PV_D.CV' != '_VLVNC_PV:OPEN' AND '//EI1_V1282B2F/PV_D.CV' != '_VLVNC_PV:OPEN')	CLOSE	FALSE	0
EI1_V1135EBN	Condition 3	Conditional Logic	(/TP01.CV' = TRUE) AND '/PV_D.CV' = '_VLVNC_PV:OPEN'	CLOSE	FALSE	5
EI1_V1150FW	Condition 1	Auto Logic	(/TP01.CV' = TRUE) AND (//EI1_LSL1150/DI1/DISC_ACT.CV' = TRUE AND '//EI1EQ_2600E/OP004_VALUE.CV' = 'NS_ENABLE:ENABLE')	OPEN	FALSE	0
EI1_V1150FW	Condition 2	Auto Logic	(/TP01.CV' = TRUE) AND (//EI1_LSL1150/DI1/DISC_ACT.CV' != TRUE OR '//EI1EQ_2600E/OP004_VALUE.CV' = 'NS_ENABLE:DISABLE')	CLOSE	FALSE	0
EI1_V1216BFY	Condition 1	Conditional Logic	/TP01.CV' = TRUE AND '//EI1EQ_2682B1/OP001_VALUE.CV' = '_DO_OFF_ON:OFF' AND '//EI1EQ_2682B2/OP001_VALUE.CV' = '_DO_OFF_ON:OFF' AND '//ESM_EIE_01/REJECT.CV' = TRUE	OPEN	FALSE	0



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EI1_V1216BFY	Condition 2	Conditional Logic	('/TP01.CV' = TRUE AND ('//EI1EQ_2682B1/OP001_VALUE.CV' = '_DO_OFF_ON:ON' OR '//EI1EQ_2682B2/OP001_VALUE.CV' = '_DO_OFF_ON:ON') OR '//ESM_EIE_01/REJECT.CV' = FALSE	CLOSE	FALSE	0
EI1_V1235ED1	Condition 1	Conditional Logic	('TP01.CV' = TRUE) AND '//EI1EQ_2635E/OP003_VALUE.CV' = 'NS_EIE_DRAIN:Drain to Storage'	OPEN	FALSE	0
EI1_V1235ED1	Condition 2	Conditional Logic	('TP01.CV' = TRUE) AND '//EI1EQ_2635E/OP003_VALUE.CV' != 'NS_EIE_DRAIN:Drain to Storage'	CLOSE	FALSE	0
EI1_V1235ED2	Condition 1	Conditional Logic	('TP01.CV' = TRUE) AND '//EI1EQ_2635E/OP003_VALUE.CV' = 'NS_EIE_DRAIN:Drain to E2682B'	OPEN	FALSE	0
EI1_V1235ED2	Condition 2	Conditional Logic	('TP01.CV' = TRUE) AND '//EI1EQ_2635E/OP003_VALUE.CV' != 'NS_EIE_DRAIN:Drain to E2682B'	CLOSE	FALSE	0
EI1_V1235EF1	Condition 1	Conditional Logic	('TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND '//EI1EQ_2635E/OP001_VALUE.CV' = 'NS_EIE_FILL_SRC:Fill From E2682B' AND '//EI1_LSH1235E1/DI1/DISC_ACT.CV' = FALSE AND ('//EI1EQ_2682B1/COMMAND_00001/S0400/ACTIVE.CV' = TRUE OR '//EI1EQ_2682B2/COMMAND_00001/S0400/ACTIVE.CV' = TRUE))	OPEN	FALSE	0

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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1235EF1	Condition 2	Conditional Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR '//EI1EQ_2635E/OP001_VALUE.CV' != 'NS_EIE_FILL_SRC:Fill From E2682B' OR '//EI1_LSH1235E1/DI1/DISC_ACT.CV' = TRUE OR ('//EI1EQ_2682B1/COMMAND_00001/S0400/ACTIVE.CV' = FALSE AND '//EI1EQ_2682B2/COMMAND_00001/S0400/ACTIVE.CV' = FALSE)))	CLOSE	FALSE	0
EI1_V1235EF1	Condition 3	Conditional Logic	(('/TP02.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND '//EI1EQ_2635E/OP001_VALUE.CV' = 'NS_EIE_FILL_SRC:Fill From E2682B' AND ('//EI1EQ_2682B1/COMMAND_00001/S0400/ACTIVE.CV' = TRUE OR '//EI1EQ_2682B2/COMMAND_00001/S0400/ACTIVE.CV' = TRUE)))	OPEN	FALSE	0
EI1_V1235EF1	Condition 4	Conditional Logic	(('/TP02.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR '//EI1EQ_2635E/OP001_VALUE.CV' != 'NS_EIE_FILL_SRC:Fill From E2682B' OR ('//EI1EQ_2682B1/COMMAND_00001/S0400/ACTIVE.CV' = FALSE AND '//EI1EQ_2682B2/COMMAND_00001/S0400/ACTIVE.CV' = FALSE)))	CLOSE	FALSE	0
EI1_V1235EF2	Condition 1	Conditional Logic	(('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND '//EI1EQ_2635E/OP001_VALUE.CV' = 'NS_EIE_FILL_SRC:Fill From Feed' AND '//EI1_LSH1235E1/DI1/DISC_ACT.CV' = FALSE))	OPEN	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1235EF2	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR //EI1EQ_2635E/OP001_VALUE.CV' != 'NS_EIE_FILL_SRC:Fill From Feed' OR //EI1_LSH1235E1/DI1/DISC_ACT.CV' = TRUE)	CLOSE	FALSE	0
EI1_V1235EF2	Condition 3	Conditional Logic	('/TP02.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND //EI1EQ_2635E/OP001_VALUE.CV' = 'NS_EIE_FILL_SRC:Fill From Feed')	OPEN	FALSE	0
EI1_V1235EF2	Condition 4	Conditional Logic	('/TP02.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR //EI1EQ_2635E/OP001_VALUE.CV' != 'NS_EIE_FILL_SRC:Fill From Feed')	CLOSE	FALSE	0
EI1_V1235EV	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND //EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND //EI1_PT1235E1/PID1/HI_HI_ACT.CV' = TRUE	OPEN	FALSE	0
EI1_V1235EV	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND //EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR //EI1_PT1235E1/PID1/HI_HI_ACT.CV' = FALSE	CLOSE	FALSE	0
EI1_V1235EW	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND //EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND //EI1_VP1241X/PV_D.CV' = '_EDC_MTR2_PV:RUNNING' AND //EI1_V1235EV/PV_D.CV' = '_VLVNC_PV:CLOSED'	OPEN	FALSE	0
EI1_V1235EW	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND (//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR //EI1_VP1241X/PV_D.CV' != '_EDC_MTR2_PV:RUNNING' OR //EI1_V1235EV/PV_D.CV' != '_VLVNC_PV:CLOSED')	CLOSE	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1246AD	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND '//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND '//EI1_VP1241X/PV_D.CV' = '_EDC_MTR2_PV:RUNNING' AND '//EI1_V1235EV/PV_D.CV' = '_VLVNC_PV:CLOSED'	OPEN	FALSE	0
EI1_V1246AD	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR '//EI1_VP1241X/PV_D.CV' != '_EDC_MTR2_PV:RUNNING' OR '//EI1_V1235EV/PV_D.CV' != '_VLVNC_PV:CLOSED')	CLOSE	FALSE	0
EI1_V1282B1D	Condition 1	Conditional Logic	'/TP01.CV' = TRUE AND ('//EI1EQ_2635E/COMMAND_00001/S0200/ACTIVE.CV' = TRUE OR '//EI1EQ_2635E/COMMAND_00001/S0500/ACTIVE.CV' = TRUE)	OPEN	FALSE	0
EI1_V1282B1D	Condition 2	Conditional Logic	'/TP01.CV' = TRUE AND ('//EI1EQ_2635E/COMMAND_00001/S0200/ACTIVE.CV' != TRUE AND '//EI1EQ_2635E/COMMAND_00001/S0500/ACTIVE.CV' != TRUE)	CLOSE	FALSE	0
EI1_V1282B1F	Condition 1	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1_LT1282B11/AI1/HI_HI_ACT.CV' != TRUE AND '//EI1_LSH1282B11/DI1/DISC_ACT' != TRUE AND '//EI1EQ_2682B1/OP001_VALUE.CV' = 'DO_OFF_ON:ON')	OPEN	FALSE	0
EI1_V1282B1F	Condition 2	Conditional Logic	('/TP01.CV' = TRUE) AND ('//EI1_LT1282B11/AI1/HI_HI_ACT.CV' = TRUE OR '//EI1_LSH1282B11/DI1/DISC_ACT' = TRUE OR '//EI1EQ_2682B1/OP001_VALUE.CV' = 'DO_OFF_ON:OFF')	CLOSE	FALSE	0
EI1_V1282B1FY	Condition 1	Conditional Logic	'/TP01.CV' = TRUE AND '//EI1EQ_2682B1/OP001_VALUE.CV' = TRUE	OPEN	FALSE	0
EI1_V1282B1FY	Condition 2	Conditional Logic	'/TP01.CV' = TRUE AND '//EI1EQ_2682B1/OP001_VALUE.CV' = FALSE	CLOSE	FALSE	0



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1282B1V	Condition 1	Conditional Logic	'/TP01.CV' = TRUE AND '//EI1_PT1282B11/AI1/HI_HI_ACT.CV' = TRUE	OPEN	FALSE	0
EI1_V1282B1V	Condition 2	Conditional Logic	'/TP01.CV' = TRUE AND '//EI1_PT1282B11/AI1/HI_HI_ACT.CV' = FALSE	CLOSE	FALSE	0
EI1_V1282B2BN	Condition 1	Conditional Logic	'/TP01.CV' = TRUE AND '//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND ('/EI1EQ_2635E/COMMAND_00001/S0300/ACTIVE.CV' = TRUE OR '//EI1EQ_2635E/COMMAND_00001/S0600/ACTIVE.CV' = TRUE) AND '//EI1_V1282B1D/PV_D.CV' = ' _VLVNC_PV:CLOSED' AND '/PV_D.CV' = ' _VLVNC_PV:CLOSED'	OPEN	FALSE	5
EI1_V1282B2BN	Condition 2	Conditional Logic	'/TP01.CV' = TRUE AND ('/EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR ('/EI1EQ_2635E/COMMAND_00001/S0300/ACTIVE.CV' = FALSE AND '//EI1EQ_2635E/COMMAND_00001/S0600/ACTIVE.CV' = FALSE) OR '//EI1_V1282B1D/PV_D.CV' = ' _VLVNC_PV:OPEN')	CLOSE	FALSE	0
EI1_V1282B2BN	Condition 3	Conditional Logic	'/TP02.CV' = TRUE AND '//EI1EQ_2600E/OP003_VALUE.CV' = FALSE AND '//EI1_V1282B1D/PV_D.CV' = ' _VLVNC_PV:CLOSED' AND '/PV_D.CV' = ' _VLVNC_PV:CLOSED'	OPEN	FALSE	5
EI1_V1282B2BN	Condition 4	Conditional Logic	'/TP02.CV' = TRUE AND ('/EI1EQ_2600E/OP003_VALUE.CV' = TRUE OR '//EI1_V1282B1D/PV_D.CV' != ' _VLVNC_PV:CLOSED')	CLOSE	FALSE	0
EI1_V1282B2BN	Condition 5	Conditional Logic	('/TP01.CV' = TRUE OR '/TP02.CV' = TRUE) AND '/PV_D.CV' = ' _VLVNC_PV:OPEN'	CLOSE	FALSE	5



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Module	Name	Description	Expression	Force SP Value	Disable	Delay ON
EI1_V1282B2D	Condition 1	Conditional Logic	'/TP01.CV' = TRUE AND ('/EI1EQ_2635E/COMMAND_00001/S0200/ACTIVE.CV' = TRUE OR '/EI1EQ_2635E/COMMAND_00001/S0500/ACTIVE.CV' = TRUE)	OPEN	FALSE	0
EI1_V1282B2D	Condition 2	Conditional Logic	'/TP01.CV' = TRUE AND ('/EI1EQ_2635E/COMMAND_00001/S0200/ACTIVE.CV' != TRUE AND '/EI1EQ_2635E/COMMAND_00001/S0500/ACTIVE.CV' != TRUE)	CLOSE	FALSE	0
EI1_V1282B2F	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND ('/EI1_LT1282B21/AI1/HI_HI_ACT.CV' != TRUE AND '/EI1_LSH1282B2/DI1/DISC_ACT' != TRUE AND '/EI1EQ_2682B2/OP001_VALUE.CV' = '_DO_OFF_ON:ON')	OPEN	FALSE	0
EI1_V1282B2F	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND ('/EI1_LT1282B21/AI1/HI_HI_ACT.CV' = TRUE OR '/EI1_LSH1282B2/DI1/DISC_ACT' = TRUE OR '/EI1EQ_2682B2/OP001_VALUE.CV' = '_DO_OFF_ON:OFF')	CLOSE	FALSE	0
EI1_V1282B2V	Condition 1	Conditional Logic	'/TP01.CV' = TRUE AND '/EI1_PT1282B21/AI1/HI_HI_ACT.CV' = TRUE	OPEN	FALSE	0
EI1_V1282B2V	Condition 2	Conditional Logic	'/TP01.CV' = TRUE AND '/EI1_PT1282B21/AI1/HI_HI_ACT.CV' = FALSE	CLOSE	FALSE	0